MONITORING AND LOGGING WITH CLOUDWATCH

WHAT IS CLOUDWATCH

CloudWatch is a monitoring service that helps us to monitor the resources, collect the metrics, create the alarms and stores the logs.

TYPES OF MONITORING

1. **Basic monitoring**
   In basic monitoring u can track the metrics within the 5 minutes of interval

2. **Detailed monitoring**
   In detailed monitoring u get the detailed metrics within the 1 minute of interval

---

METRICS

It is the collected monitoring data of resources (like CPU utilization, network traffic, disk usage) over time.

CLOUDWATCH EVENTS

Events (CloudWatch Events / EventBridge): Notifications generated when a change or action happens in AWS (like instance start/stop, login, etc.).

CLOUDWATCH ALARM STATE

1. **OK:** Metric is within the defined threshold (everything is normal).
2. **ALARM:** Metric has crossed the threshold (issue detected).
3. **INSUFFICIENT_DATA:** Not enough data available to determine the state.

PRACTICAL ON CLOUDWATCH WITH SNS

1.create a instance

1. name ------ ubuntu
2. machine --- ubuntu
3. sg -------- http/ssh/
4. create a instance

2.create a SNS topic

1. create a topic
2. type ----- standard
3. name ----- increase-traffic
4. create a topic

3.create a subscription in SNS

1. create a subscription
2. topic arn ----> increase-traffic
3. protocol -----> email
4. create subscription and go to ur gmail and confirm subscription

4.create a CloudWatch alarm

1. create a alarm
2. select a metric ---> ec2 ----> per-instance metric
3. select instance ---> select CPU-utilization
4. period --- 1 minute
5. threshold type --- static
6. utilization ------- greater/equal
7. than ---------- 50%
8. select existing topic --- increase-traffic
9. alarm name -------- increase-alarm
10. create a alarm

5.Connect Instance & Generate Load

1. apt update
2. apt install stress -y
3. stress --cpu 4 --timeout 300s
